
Verify-or-Abstain Evidence Pipelines: When Cryptographic Custody Is Necessary but Not Sufficient

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Scientific evidence pipelines increasingly rely on content-addressed custody and
2 hash-valid artifacts, yet syntactically valid objects can remain semantically in-
3 consistent. We present an evidence-pipeline reliability study centered on explicit
4 terminal states, row-level accounting invariants, and closed-vocabulary identity
5 guards. Preregistered audits show that (i) a historical aggregate reported zero ab-
6 stentions while row-level data contained three; (ii) a successor implementation
7 with derived aggregates restores invariant consistency; (iii) a deterministic iden-
8 tity contract passes under synthetic fixtures but a single observed-source pair ad-
9 mits downstream passage when a guard is bypassed; and (iv) historical sequence
10 and copy-number corpora could not be recovered from admitted provenance. A
11 morphology null benchmark is retained as disclosed negative cached evidence.
12 We do not claim therapeutic efficacy, clinical utility, population-level biomedical
13 error rates, or real-world evidence.

14 1 Introduction

15 Machine-learning systems for biomedical hypothesis generation must separate *what was computed*
16 from *what may be claimed*. Content-addressed storage and hash verification establish integrity of
17 bytes, not meaning: aggregates can disagree with row ledgers, evaluation wiring can fail to exercise
18 intended disagreements, and missing corpora can leave historical numbers unverified. We frame this
19 work as an evidence-pipeline control problem: normalize, validate, admit or abstain, derive results,
20 and attach claim ceilings before projection to manuscripts [? ?].

21 2 Related Work and Terminology

22 Selective prediction and reject options allow models to abstain under uncertainty [? ?]. Workflow
23 and data provenance standards document lineage without guaranteeing semantic correctness [?].
24 Biomedical identifier normalization relies on authorities such as HGNC and UniProt [? ?]. Re-
25 producibility guidance distinguishes replication from reproducibility and encourages reporting null
26 results [?]. Regulatory Real-World Evidence frameworks define boundaries we do not claim here
27 [?].

28 We distinguish *cryptographic custody* (hash-valid, content-addressed storage) from *semantic verifi-*
29 *cation* (terminal-state invariants, identity resolution, aggregate-row consistency).

30 **Synthetic fixture.** An intentionally constructed input used to test deterministic software behavior.

31 **Canonical contract fixture.** A project-defined synthetic or curated fixture whose expected output
32 is fixed before execution and used to test a deterministic contract.

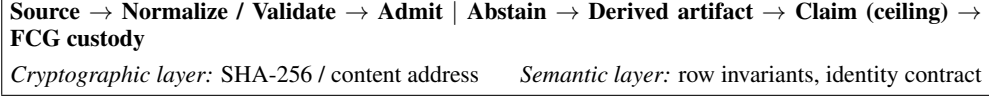


Figure 1: Verify-or-abstain evidence pipeline. Custody without semantic checks can admit inconsistent aggregates.

Experiment	Unit	Guard	Condition	N	Ceiling
GAP accounting repair	row	aggregate invariant	successor run	4	semantic audit
G3 contract	fixture	identity	Lane A	synth.	deterministic
G3 source ablation	pair	identity bypass	Lane B	1	descriptive
G1 provenance	corpus	—	not recovered	—	audit only
G2 provenance	corpus	—	not recovered	—	audit only
Morphology null	ranking	shuffle null	cached	50	repurposing hyp.

Table 1: Primary results from frozen experiment artifacts. Historical G1/G2 counts are excluded.

Observed source record. An actual record retrieved from an identified external source rather than invented for testing. Evaluations using such records are *observed-source evaluations*; they are not Real-World Evidence (RWE) unless regulatory RWD/RWE criteria are satisfied [?].

External biological evidence. A source-backed biological record or publication used as evidence for a biological assertion.

Abstention / reject option. Following Chow [?] and selective-prediction literature [?], a pipeline may emit a terminal ABSTAIN state rather than admit an under-supported record.

Closed-vocabulary identity resolution. Map source strings to a frozen alias table with exact, mapped, or unresolved outcomes; unresolved inputs must not silently pass as canonical.

Provenance. Origin and processing lineage of entities [?]. Hash equality does not imply semantic consistency between derived artifacts.

3 Verify-or-Abstain Architecture

Figure 1 sketches the pipeline: SOURCE → normalize/validate → admit or abstain → derived result → claim with ceiling → frozen custody graph. Semantic verification (terminal states, invariants, identity guards) complements cryptographic custody.

4 Experimental Design

Experiments are preregistered where noted; August 13 historical runs are preserved and not retroactively relabeled. EXP-GAP-ACCOUNTING-001.1 repairs aggregate derivation from rows. EXP-004 ablates a closed-vocabulary identity guard (Lane A: contract; Lane B: $N=1$ observed-source pair). EXP-002/003 provenance audits and external recovery (EXP-002-PROV.1, EXP-003-PROV.1) search admitted custody without rerunning frozen closeouts. EXP-005 preregistration remains locked but corpus/metric incomplete (not executed). EXP-006 documents a cached morphology null benchmark.

5 Results

Table 1 summarizes terminal classifications and claim ceilings from machine-readable receipts.

GAP accounting. Historical summary reported zero abstentions; row-level candidates contained three ABSTAIN terminals. Successor run: $N_{input}=4$, $N_{admitted}=1$, $N_{abstained}=3$, summary abstain count = 3; invariants pass.

Identity guard. Lane A: deterministic closed-vocabulary contract passes. Lane B: one observed source/canonical pair (TAZ/cardioliipin module → TAZ) yields ABSTAIN under guard

63 and downstream admission under bypass (*unsupported identity admission*); not a population error
64 rate. Historical evaluation invoked `reconcile(symbol, symbol)`; regression confirms the wiring
65 defect.

66 **Provenance recovery.** Admitted external bootstrap package (206 MB, SHA-256 verified) does not
67 contain recoverable G1/G2 derivation chains; numeric substring matches in unrelated CSV fields
68 are rejected as evidence.

69 **Null benchmark.** Cached evaluation: known-pair rank 28/50, reciprocal-rank enrichment vs. shuf-
70 fle < 1 ; disclosed miss not suppressed.

71 6 Discussion and Limitations

72 Scope is intentionally small: $N=1$ observed-source ablation; G1/G2 inputs not recovered; same
73 designers performed parts of evaluation; no therapeutic efficacy, biological rescue, clinical utility,
74 RWE, or population prevalence claims; EXP-005 replication not executed pending corpus freeze;
75 infrastructure integrity does not establish semantic correctness.

76 7 Reproducibility and Provenance

77 Artifacts, preregistrations, and SHA-256 manifests reside in the project Frozen Custody Graph.
78 Anonymous supplementary materials omit identifying metadata. Historical unverified counts remain
79 quarantined as UNVERIFIED_HISTORICAL_CLAIM.

80 8 Conclusion

81 Hash-valid custody is necessary but insufficient for trustworthy evidence pipelines. Explicit termi-
82 nal states and invariants surface failure modes that remain syntactically admissible—a prerequisite
83 before any biomedical performance claim.

84 References

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